

GATE CS/IT: Computer Networks

IP Address

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What is a Computer Network?

- **Definition:** A system of interconnected, autonomous computing devices designed to exchange data and share resources.
- **Core Components:**
 - **Nodes:** End systems (hosts, servers) and intermediate devices (routers, switches, hubs).
 - **Links:** Transmission media (copper wire, optical fiber, wireless channels) that connect nodes.
- **Primary Objectives:**
 - Resource sharing (data, hardware, software).
 - High reliability (redundancy and fault tolerance).
 - Interprocess communication across distributed systems.

Key Network Metrics (GATE Focus)

- **Performance Criteria:** Essential for solving GATE numericals.
- **Throughput:** The actual rate at which data is successfully transferred over a link (measured in bps).
- **Bandwidth:** The maximum data capacity of a link.
- **Total Delay (Latency):** The time taken for a packet to travel from source to destination.

$$T_{total} = T_{transmission} + T_{propagation} + T_{queuing} + T_{processing}$$

- **Reference Models:** The foundation of Computer Networks study relies on layered architectures:
 - **OSI Model** (7 Layers)
 - **TCP/IP Model** (4 or 5 Layers)

Introduction to IP Addressing

- **Definition:** A logical, numeric address assigned to every device connected to a computer network that uses the Internet Protocol.
- **OSI Layer:** Operates at the **Network Layer** (Layer 3).
- **Purpose:**
 - **Host identification:** Uniquely identifies a specific device on a network.
 - **Network addressing (Routing):** Determines the path to forward packets across multiple networks.
- **Primary Versions:**
 - **IPv4:** 32-bit address space, supporting approx. 2^{32} (4.3 billion) addresses.
 - **IPv6:** 128-bit address space, supporting approx. 2^{128} addresses (designed to solve IPv4 exhaustion).

IPv4 Address Structure (GATE Focus)

- **Format:** 32 bits long, divided into 4 octets (8 bits each). Typically represented in **Dotted-Decimal Notation** (e.g., 192.168.1.15).
- **Octet Range:** Each octet ranges from 0 to 255.
- **Hierarchical Design:** An IP address is strictly divided into two parts:
 - ① **Network ID (NetID):** Identifies the specific physical network. Routers use this part to forward packets globally.
 - ② **Host ID (HostID):** Identifies the specific node/device within that local network.
- **Addressing Architectures:**
 - **Classful Addressing:** Fixed boundary for NetID and HostID (Classes A, B, C, D, E).
 - **Classless Inter-Domain Routing (CIDR):** Variable boundary, represented as IP/Prefix (e.g., 10.0.0.0/24). **Most numericals in GATE come from CIDR and Subnetting!**

Classful Addressing: The Prefix Approach

- **Concept:** The 32-bit IPv4 address space is divided into five classes (A, B, C, D, E) based on fixed **leading bits** in the first octet.
- **Fixed Boundaries:** The boundary between the Network ID and Host ID is fixed for each class.
- **Subnet Masks (Default):**
 - Class A: 255.0.0.0 (8 bits for Network)
 - Class B: 255.255.0.0 (16 bits for Network)
 - Class C: 255.255.255.0 (24 bits for Network)
- **Host Subtraction Rule:** In any class, the valid number of hosts per network is always $(2^{\text{Host Bits}} - 2)$.
 - All 0s in Host ID = Network Address.
 - All 1s in Host ID = Direct Broadcast Address (DBA).

Deriving Class A

- **Leading Bit:** Always 0.
- **Structure:** 8 bits Network ID, 24 bits Host ID.
- **First Octet Range:** 00000000_2 to $01111111_2 \implies 0$ to 127 .
 - **Valid Range: 1 to 126** (Since 0 is used for default routing and 127 is reserved for Loopback/Localhost testing).
- **GATE Formulas for Class A:**
 - **Total Networks:** $2^{(8-1)} - 2 = 2^7 - 2 = 126$ valid networks. (We subtract 1 bit for the fixed '0', and subtract 2 for the special networks 0 and 127).
 - **Total Hosts per Network:** $2^{24} - 2 = 16,777,214$ hosts.
- **Use Case:** Huge networks with millions of hosts.

Deriving Class B

- **Leading Bits:** Always **10**.
- **Structure:** 16 bits Network ID, 16 bits Host ID.
- **First Octet Range:** 10000000_2 to $10111111_2 \implies$ **128 to 191**.
- **GATE Formulas for Class B:**
 - **Total Networks:** $2^{(16-2)} = 2^{14} = 16,384$ networks. (We subtract 2 bits for the fixed '10'). No networks are subtracted here.
 - **Total Hosts per Network:** $2^{16} - 2 = 65,534$ hosts.
- **Use Case:** Medium to large-sized organizations.

Deriving Class C

- **Leading Bits:** Always **110**.
- **Structure:** 24 bits Network ID, 8 bits Host ID.
- **First Octet Range:** 11000000_2 to $11011111_2 \implies$ **192 to 223**.
- **GATE Formulas for Class C:**
 - **Total Networks:** $2^{(24-3)} = 2^{21} = 2,097,152$ networks. (Subtract 3 bits for the fixed '110').
 - **Total Hosts per Network:** $2^8 - 2 = 254$ hosts.
- **Use Case:** Small organizations and local area networks.

- **Class D (Multicasting):**

- **Leading Bits:** 1110.
- **First Octet Range:** 11100000_2 to $11101111_2 \implies 224$ to 239 .
- **Concept:** No concept of Network ID or Host ID. A Class D address represents a *group* of hosts rather than a specific network/host. Used in protocols like OSPF, RIPv2.

- **Class E (Experimental/Reserved):**

- **Leading Bits:** 1111.
- **First Octet Range:** 11110000_2 to $11111111_2 \implies 240$ to 255 .
- **Concept:** Reserved for future use and R&D. (Note: 255.255.255.255 is used as the Limited Broadcast Address).

Unicasting: One-to-One Communication

- **Concept:** The transmission of a packet from a single source to a **single, specific destination**.
- **Addressing Rules:**
 - **Network Layer (IP):** The destination IP address identifies a unique host (e.g., sending a packet to 192.168.10.50).
 - **Data Link Layer (MAC):** The Least Significant Bit (LSB) of the first byte of the destination MAC address is always **0** for unicast.
- **Behavior:**
 - Switches forward the frame only to the port where the target MAC address is learned.
 - Routers forward the packet toward the specific destination network using routing tables.
- **Typical Applications:** Web browsing (HTTP/HTTPS), File transfers (FTP), Email (SMTP), SSH.

Broadcasting: One-to-All Communication

- **Concept:** The transmission of a packet from a single source to **all nodes** within a network domain.
- **Data Link Layer:** The destination MAC address is FF:FF:FF:FF:FF:FF (all 48 bits are set to 1). Switches flood this frame out of all ports.
- **Types of IP Broadcast (Crucial for GATE):**
 - 1 **Limited Broadcast Address (LBA):**
 - IP Address: 255.255.255.255 (All 32 bits are 1).
 - Reaches all hosts on the **local network** only.
 - **Golden Rule:** A router *never* forwards a limited broadcast packet. It acts as a boundary for broadcast domains.
 - 2 **Directed/Direct Broadcast Address (DBA):**
 - IP Format: Target Network ID + All 1s in the Host ID (e.g., 192.168.1.255 for a Class C network).
 - Reaches all hosts on a **specific remote network**.
 - Routers *can* route this packet to the destination network's final router, which then broadcasts it locally (though modern routers disable this by default to prevent Smurf attacks).

Motivation for Subnetting: The Address Wastage Problem

- **Inflexible Classful Architecture:** Classful addressing forces networks into fixed sizes (Class A, B, or C), which rarely matches an organization's actual needs.
- **Massive Wastage of IP Addresses:**
 - Example: An organization needing 500 IP addresses cannot use Class C (only 254 hosts). It must be assigned a Class B network.
 - Class B provides 65,534 hosts. Assigning it for 500 hosts wastes $\approx 65,034$ valid IP addresses!
 - This inefficiency heavily accelerated **IPv4 Address Exhaustion**.
- **Broadcast Storms:**
 - In a single flat network (like a full Class B), a broadcast message is flooded to all 65,534 hosts.
 - This causes severe CPU interruptions, consumes immense bandwidth, and drastically reduces overall network performance.

The Solution: Concept and Benefits of Subnetting

- **Definition:** The process of logically dividing a single large contiguous IP network into multiple smaller, manageable sub-networks (subnets).
- **The Mechanism (GATE Focus):**
 - We borrow n contiguous bits from the **Host ID** portion to create a **Subnet ID**.
 - New 3-level hierarchy: **Network ID + Subnet ID + Host ID**.
- **Key Advantages:**
 - 1 **Reduced Broadcast Domain:** Routers separate subnets and do not forward limited broadcasts, confining broadcast traffic to small groups.
 - 2 **Efficient IP Allocation:** Prevents address wastage by tailoring subnet sizes to exact requirements (leading to Variable Length Subnet Masking - VLSM).
 - 3 **Enhanced Security:** Traffic moving between different subnets must pass through a router or firewall, allowing access control policies to be enforced.

Example: Subnetting a Class C Network

- **Given Network:** 200.1.5.0 (Class C)
- **Default Subnet Mask:** 255.255.255.0 or /24 (24 Net bits, 8 Host bits).
- **Objective:** Divide this network into **2 equal subnets**.
- **Subnetting Math (GATE Approach):**
 - To create 2 subnets, borrow n bits from the Host ID where $2^n \geq 2$.
 - Therefore, $n = 1$ bit borrowed.
 - **New Subnet Mask:** 24 (default) + 1 (borrowed) = /25.
 - In decimal, the last octet mask is $10000000_2 = 128$. Full mask: 255.255.255.128.
 - **Remaining Host Bits:** $8 - 1 = 7$ bits.
 - **Valid Hosts per Subnet:** $2^7 - 2 = 126$ hosts.

Calculating Net ID and DBA for the Subnets

- The last octet has 1 Subnet Bit and 7 Host Bits: [S H H H H H H H]
- **Subnet 1 (Subnet Bit = 0):**
 - **Network ID (All Host bits 0):** $00000000_2 = 0 \implies 200.1.5.0/25$
 - **Directed Broadcast Address (All Host bits 1):**
 $01111111_2 = 127 \implies 200.1.5.127$
 - *Valid Host Range:* 200.1.5.1 to 200.1.5.126
- **Subnet 2 (Subnet Bit = 1):**
 - **Network ID (All Host bits 0):**
 $10000000_2 = 128 \implies 200.1.5.128/25$
 - **Directed Broadcast Address (All Host bits 1):**
 $11111111_2 = 255 \implies 200.1.5.255$
 - *Valid Host Range:* 200.1.5.129 to 200.1.5.254
- **GATE Trick (Block Size):** Block size = $256 - \text{Subnet Mask value}$. Here, $256 - 128 = 128$. Subnet Net IDs will always be multiples of the block size (0, 128).

Concept of a Subnet Mask

- **Definition:** A 32-bit value used to clearly distinguish the Network (and Subnet) portion from the Host portion of an IP address.
- **The Golden Rule:** A valid subnet mask must consist of **continuous 1s followed by continuous 0s**.
 - Example of valid mask: 11111111.11111111.11111111.10000000 (255.255.255.128)
 - Example of invalid mask: 255.255.255.10 (Binary is 00001010 - 1s are not continuous)
- **Logical Mapping:**
 - 1s represent the **Network ID + Subnet ID**.
 - 0s represent the **Host ID**.
- **Representations:**
 - **Dotted-Decimal:** 255.255.255.224
 - **CIDR Notation (Slash):** /27 (indicating twenty-seven consecutive 1s).

How Routers Use the Subnet Mask

- **The Mechanism:** When a router receives a packet, it must determine the target network. It performs a **Bitwise AND** operation between the destination IP address and the Subnet Mask.

- **Mathematical Rule:**

$$\text{IP Address AND Subnet Mask} = \text{Network ID}$$

- **GATE Example Calculation:**

- **Destination IP:** 140.20.15.135
- **Subnet Mask:** 255.255.255.192 (or /26)
- We only need to calculate the octet where the mask is not 255 or 0 (the 4th octet).

- **Bitwise AND on the 4th octet:**

IP (135): 10000111₂

Mask (192): 11000000₂

AND Result: **10000000₂ $\Rightarrow$ 128**

- **Resulting Subnet ID:** 140.20.15.128

Concept of a Routing Table

- **Definition:** A data table stored in a router (or networked computer) that lists the routes to particular network destinations.
- **Purpose:** When a router receives a packet, it searches this table for the longest prefix match to decide where to send the packet next.
- **Key Components (GATE Focus):**
 - **Destination Network (Net ID):** The target IP subnet.
 - **Subnet Mask:** Used to extract the Network ID from the incoming packet's destination IP address.
 - **Next Hop (Gateway):** The IP address of the next router in the path. If the network is directly connected, there is no next hop.
 - **Interface:** The physical/logical port (e.g., eth0, Fa0/1) through which the packet leaves the router.

Scenario Setup: 4 Subnets in Class C

- **Base Network:** 192.168.10.0 /24 (Class C)
- **Requirement:** Divide into 4 equal subnets.
- **Subnetting Math:**
 - Bits borrowed: $2^n \geq 4 \implies n = 2$.
 - New Subnet Mask: /24 + 2 = /**26** (255.255.255.192).
 - Block Size: 256 - 192 = **64**.
- **The 4 Subnets Generated:**
 - ① **Subnet 0:** 192.168.10.0/26
 - ② **Subnet 1:** 192.168.10.64/26
 - ③ **Subnet 2:** 192.168.10.128/26
 - ④ **Subnet 3:** 192.168.10.192/26

Routing Table for the Internal Router

- Assume a central router connects all 4 subnets.
- The router's interfaces (eth0 to eth3) are assigned the **first valid IP** of each subnet to act as the default gateway for hosts in that subnet.

Destination Network	Subnet Mask	Next Hop	Interface	Router IP on Interface
192.168.10.0	255.255.255.192	Directly Connected	eth0	192.168.10.1
192.168.10.64	255.255.255.192	Directly Connected	eth1	192.168.10.65
192.168.10.128	255.255.255.192	Directly Connected	eth2	192.168.10.129
192.168.10.192	255.255.255.192	Directly Connected	eth3	192.168.10.193
0.0.0.0 (Default)	0.0.0.0	Router_ISP_IP	eth4	Outside Interface

Table: Routing table showing directly connected routes.

Variable Length Subnet Masking (VLSM)

- **The Problem with Fixed Subnetting (FLSM):** Subnetting a network into equal pieces often leads to wastage if different departments need drastically different numbers of IP addresses.
- **Definition of VLSM:** Subnetting a subnet. It allows a single network space to be divided into unequal sub-networks with different subnet masks.
- **Golden Rules for VLSM (GATE Tips):**
 - ① **Order matters:** *Always* allocate subnets starting from the largest requirement down to the smallest requirement.
 - ② **Continuous Allocation:** The next subnet must start at the very next available IP address.
 - ③ **Alignment Rule:** The Network ID of a subnet must be a multiple of its block size.

VLSM Example: Class C divided into 128, 64, 64

- **Base Network:** 192.168.1.0 /24 (Total block size = 256)
- **Requirement:** Divide into 3 subnets of block sizes 128, 64, and 64.
- **Step 1: Largest Subnet (Block Size = 128)**
 - Formula: $2^h \geq 128 \implies$ Host bits (h) = 7.
 - **Subnet Mask:** $32 - 7 =$ /25 (or 255.255.255.128).
 - **Network ID:** Starts at **192.168.1.0/25**
 - **Range:** 192.168.1.0 to 192.168.1.127
 - **DBA: 192.168.1.127**
 - *Valid Hosts:* $128 - 2 = 126$.

VLSM Example: Allocating the 64-Block Subnets

• Step 2: Second Subnet (Block Size = 64)

- Next available IP starts at 192.168.1.128.
- Formula: $2^h \geq 64 \implies$ Host bits (h) = 6.
- **Subnet Mask:** $32 - 6 = /26$ (or 255.255.255.192).
- **Network ID:** 192.168.1.128/26
- **Range:** 192.168.1.128 to 192.168.1.191
- **DBA:** 192.168.1.191

• Step 3: Third Subnet (Block Size = 64)

- Next available IP starts at 192.168.1.192.
- Formula: $2^h \geq 64 \implies$ Host bits (h) = 6.
- **Subnet Mask:** /26.
- **Network ID:** 192.168.1.192/26
- **Range:** 192.168.1.192 to 192.168.1.255
- **DBA:** 192.168.1.255

Mastering Subnetting: The Core Formulas

- To find subnets and hosts, compare the **Given Subnet Mask** with the **Default Class Mask**.
- **Step 1: Find the Default Mask**
 - Class A: /8
 - Class B: /16
 - Class C: /24
- **Step 2: Calculate Borrowed Bits (n)**
 - $n = (\text{Bits in Given Mask}) - (\text{Bits in Default Mask})$
 - **Total Subnets** = 2^n (Note: Modern GATE standard uses 2^n . Only use $2^n - 2$ if explicitly asked for "usable subnets per legacy RFC 950").
- **Step 3: Calculate Host Bits (h)**
 - $h = 32 - (\text{Bits in Given Mask})$
 - **Valid Hosts per Subnet** = $2^h - 2$

Problem Set 1: Class C and Class B

Problem 1: Class C Network

- **Given:** Class C, Subnet Mask = 255.255.255.224
- **Analysis:** The mask in binary ends in 11100000, which is /27.
- **Borrowed bits (n):** $27 - 24(\text{default}) = 3$ bits.
- **Host bits (h):** $32 - 27 = 5$ bits.
- **Result:**
 - Total Subnets = $2^3 = 8$ subnets.
 - Hosts per Subnet = $2^5 - 2 = 32 - 2 = 30$ valid hosts.

Problem 2: Class B Network

- **Given:** Class B, Subnet Mask = 255.255.248.0
- **Analysis:** The 3rd octet is 248 (11111000_2). Total mask is $8 + 8 + 5 = /21$.
- **Borrowed bits (n):** $21 - 16(\text{default}) = 5$ bits.
- **Host bits (h):** $32 - 21 = 11$ bits.
- **Result:** Total Subnets = $2^5 = 32$, Hosts = $2^{11} - 2 = 2046$.

Problem Set 2: Class A and the Point-to-Point Edge Case

Problem 3: Class A Network

- **Given:** Class A, Subnet Mask = 255.192.0.0
- **Analysis:** The 2nd octet is 192 (11000000_2). Total mask is $8 + 2 = /10$.
- **Borrowed bits (n):** $10 - 8(\text{default}) = 2$ bits.
- **Host bits (h):** $32 - 10 = 22$ bits.
- **Result:**
 - Total Subnets = $2^2 = 4$ subnets.
 - Hosts per Subnet = $2^{22} - 2 = 4,194,304 - 2 = 4,194,302$ hosts.

Problem 4: The /30 Subnet (Very Common in GATE)

- **Given:** Class C, Subnet Mask = 255.255.255.252 (/30)
- **Borrowed bits (n):** $30 - 24 = 6$ bits. Total Subnets = $2^6 = 64$.
- **Host bits (h):** $32 - 30 = 2$ bits.
- **Hosts per Subnet:** $2^2 - 2 = 4 - 2 = 2$ valid hosts.
- **GATE Insight:** A /30 subnet is universally used for Point-to-Point router connections, exactly because it provides exactly 2 IPs (one for each router) with zero wastage!

Motivation for CIDR: The Crises of the 1990s

- **Rapid IPv4 Exhaustion (The Class B Problem):**

- An organization needing 2,000 IPs could not use Class C (only 254 hosts). It had to be assigned a Class B network (65,534 hosts).
- This resulted in massive IP wastage, rapidly depleting the available unallocated Class B addresses.

- **Routing Table Explosion:**

- To avoid Class B wastage, ISPs started giving multiple Class C networks to a single organization (e.g., eight contiguous Class Cs).
- Each Class C network required its own separate entry in the core internet routing tables, causing extreme router memory consumption and slowed routing lookups.

- **The Fix:** A classless system that allows variable-sized network allocations and **Route Aggregation** (Supernetting) to shrink routing tables.

Definitions: Classless Inter-Domain Routing (CIDR)

- **Definition:** An IP addressing architecture that eliminates the strict, fixed boundaries of Class A, B, and C networks. Networks are defined by an arbitrary prefix length.
- **CIDR Notation (Slash Notation):**
 - Represented as `IP_Address / n`
 - n = The **Network Prefix Length** (number of contiguous 1s in the subnet mask).
 - Example: `192.168.10.32/28` means the first 28 bits identify the network portion, leaving $32 - 28 = 4$ bits for the host portion.
- **Classless Paradigm:** In CIDR, a router completely ignores the leading bits (0, 10, 110). It routes packets based purely on the provided prefix length (n) and the longest prefix match algorithm.

Rules for Forming a Valid CIDR Block (GATE Focus)

- When an Internet Service Provider (ISP) allocates a block of IP addresses using CIDR, **three strict mathematical rules** must apply:
 - ① **Contiguous Allocation:** All IP addresses in the allocated block must be strictly consecutive.
 - ② **Power of 2 Block Size:** The total number of IP addresses in the block must be exactly a power of 2.
 - Block Size = 2^h , where $h = 32 - n$.
 - ③ **Alignment Rule (Most Tested in GATE):** The first IP address of the block (the Network ID) must be evenly divisible by the block size.
 - **Binary verification:** If the block size is 2^h , the last h bits of the first IP address in the block **MUST** be exactly all 0s.

Subnetting in CIDR: The Mathematical Model

- **Concept:** In CIDR, we don't start from a default classful mask (/8, /16, or /24). We start from an arbitrary allocated block with prefix /n.
- **The Subnetting Rule:** To divide a CIDR block of /n into multiple subnets, we borrow k **bits** from the remaining host bits.
- **GATE Formulas for CIDR Subnetting:**
 - **New Subnet Prefix:** $/(n + k)$
 - **Total Number of Subnets created:** 2^k
 - **Total IPs per Subnet:** $2^{32-(n+k)}$
 - **Valid Hosts per Subnet:** $2^{32-(n+k)} - 2$
- **Key Insight:** Every time you add 1 to the prefix length (e.g., from /20 to /21), you exactly **halve** the number of IPs in the block and double the number of subnets.

GATE Numerical: Equal Subnetting in CIDR

- **Problem:** An ISP allocates the block 100.1.16.0 /20 to an organization. The organization wants to divide it into **8 equal subnets**. Find the new prefix and the range of the first two subnets.
- **Step 1: Calculate borrowed bits (k)**
 - $2^k \geq 8 \implies k = 3$ bits.
- **Step 2: Calculate New Prefix and Block Size**
 - New Prefix = $/20 + 3 = /23$.
 - IPs per subnet = $2^{32-23} = 2^9 = 512$ IPs.
- **Step 3: Determine the Subnet Ranges**
 - 512 IPs span exactly **2 complete Class C blocks** (or a step of 2 in the 3rd octet).
 - **Subnet 0:** 100.1.16.0/23 to 100.1.17.255
 - **Subnet 1:** 100.1.18.0/23 to 100.1.19.255
 - *(Notice how the 3rd octet increments by 2 for each new subnet's Network ID: 16, 18, 20...)*

Advanced GATE Concept: CIDR VLSM Allocation

- **Scenario:** You are given a /24 block and need subnets for 100 hosts, 50 hosts, and 12 hosts.
- **Rule:** Always allocate the largest block first to satisfy the alignment rule.
- **Allocation Steps:**
 - ① **100 hosts:** Needs $2^7 = 128$ block size.
 - Prefix: $32 - 7 = /25$.
 - Subnet 1: IP.0 /25 (Range: 0 to 127).
 - ② **50 hosts:** Needs $2^6 = 64$ block size.
 - Prefix: $32 - 6 = /26$.
 - Subnet 2 starts at the next available IP: IP.128 /26 (Range: 128 to 191).
 - ③ **12 hosts:** Needs $2^4 = 16$ block size.
 - Prefix: $32 - 4 = /28$.
 - Subnet 3 starts at next IP: IP.192 /28 (Range: 192 to 207).
- **Unallocated IPs:** The range IP.208 to IP.255 remains available for future subnets!

Practical 1: Inspecting IP and Subnet Mask

- **Command:** `ipconfig` (or `ipconfig /all` for MAC/DNS details)
- **What it shows:** Your host's assigned IPv4 Address, Subnet Mask, and Default Gateway.
- **Connecting to Theory:**
 - **IPv4 Address:** e.g., 192.168.1.45
 - **Subnet Mask:** e.g., 255.255.255.0. You can immediately deduce this is a /24 CIDR network.
 - **Network ID Calculation:** Perform a mental bitwise AND operation. $192.168.1.45 \text{ AND } 255.255.255.0 = 192.168.1.0$.
 - **Default Gateway:** e.g., 192.168.1.1. The IP of the router interface directly connected to your subnet. Any packet not meant for the 192.168.1.0/24 block is forwarded here.

Practical 2: Analyzing the Host Routing Table

- **Command:** `route print -4` (or `netstat -rn`)
- **Concept:** Even end-hosts (your laptop) have routing tables to decide if a packet stays on the local network or goes to the gateway.
- **Mapping Output Columns to GATE Concepts:**
 - **Network Destination:** The target Subnet ID. The 0.0.0.0 entry represents the default route (matches everything else).
 - **Netmask:** The Subnet Mask used by the OS for the Bitwise AND operation against the target IP.
 - **Gateway (Next Hop):** If the destination is within your own subnet, Windows displays On-link (Directly Connected). Otherwise, it displays the Router's IP.
 - **Metric:** If multiple routes match, the one with the lowest cost/metric is chosen.
- *Note: The OS strictly applies the **Longest Prefix Match** algorithm here.*

Practical 3: Verifying Connectivity and Hops

- **Command:** `ping <IP_Address>`
- **Concept (Unicasting):** Sends ICMP Echo Request packets to a specific host. Verifies two-way Layer 3 connectivity.
- **Concept (Broadcasting):** If you ping your subnet's Directed Broadcast Address (e.g., 192.168.1.255), you are testing the DBA concept. (Note: Modern Windows firewalls often block responding to broadcast pings for security).
- **Command:** `tracert <Domain_Name_or_IP>`
- **Concept (Routing Path):** Visualizes the Next-Hop routing process across the Internet.
- **How it works:** It manipulates the **Time To Live (TTL)** field in the IP header. Each line printed in the command prompt represents a different router (hop) that processed your packet using its own internal routing table.

Why Does a Sender Need a Subnet Mask?

- **The Core Problem:** When a host wants to send a packet to a destination IP, it must first answer one critical question: *"Is the destination on my local network, or on a remote network?"*
- **The Role of the Subnet Mask:** The sender uses its **own subnet mask** to make this crucial local vs. remote routing decision.
- **Outcome of the Decision:**
 - **Local Network:** The sender can deliver the frame directly. It uses ARP (Address Resolution Protocol) to find the MAC address of the *destination host*.
 - **Remote Network:** The sender cannot deliver directly. It must forward the packet to the **Default Gateway (Router)**. It uses ARP to find the MAC address of the *router's interface*.

The Mathematical Mechanism (GATE Focus)

- To determine the location of the destination, the sender's Operating System performs two bitwise AND operations.
- **Step 1: Calculate Own Network ID**

$$\text{Sender_Net_ID} = (\text{Sender IP}) \text{ AND } (\text{Sender Subnet Mask})$$

- **Step 2: Calculate Apparent Target Network ID**

$$\text{Target_Net_ID} = (\text{Destination IP}) \text{ AND } (\text{Sender Subnet Mask})$$

- **Step 3: The Comparison**

- If **Sender_Net_ID == Target_Net_ID**: The destination is local. Frame is built with the Destination MAC.
- If **Sender_Net_ID $\neq$ Target_Net_ID**: The destination is remote. Frame is built with the Gateway's MAC.
- *Crucial Insight*: The sender **does not know** (and doesn't need to know) the destination's actual subnet mask. It only applies its *own* mask to see if the destination falls within its own logical boundary.

Concept of Supernetting (Route Aggregation)

- **Definition:** The process of combining multiple, smaller contiguous networks into a single, larger network (a supernet). It is the exact mathematical inverse of subnetting.
- **Primary Objective:** To minimize the size of routing tables in core internet routers (also known as Route Summarization).
- **How it Works:** Instead of a router having 4 separate entries for 4 different Class C networks, it only needs 1 entry for the summarized supernet.
- **Prefix Behavior:**
 - **Subnetting:** Moves the boundary to the *right*. Borrow bits from Host to Network. Prefix length increases (e.g., /24 to /26).
 - **Supernetting:** Moves the boundary to the *left*. Borrow bits from Network to Host. Prefix length decreases (e.g., /24 to /22).

The 3 Golden Rules for Supernetting

- To successfully aggregate multiple network blocks into a single valid CIDR supernet, **all three** of these conditions must be met:
 - ① **Contiguous Rule:** All the IP networks being combined must be strictly continuous (e.g., 10.0.1.0, 10.0.2.0, 10.0.3.0).
 - ② **Power of 2 Rule:** The total number of networks being combined must be exactly a power of 2 ($N = 2, 4, 8, 16, \dots$).
 - ③ **Alignment Rule (Crucial for GATE):** The first (lowest) Network ID in the contiguous block must be evenly divisible by the total number of blocks being combined (N).
 - *Practical Check:* If you are combining N Class C networks, look at the 3rd octet of the very first network. That 3rd octet must be a multiple of N (e.g., 0, N , $2N$, $3N \dots$).

GATE Example: Validating a Supernet

- **Given Networks:** A router receives routes for 4 Class C networks:

- 200.1.4.0 /24
- 200.1.5.0 /24
- 200.1.6.0 /24
- 200.1.7.0 /24

- **Applying the Rules:**

- 1 Contiguous? Yes (4, 5, 6, 7).
- 2 Power of 2? Yes ($4 = 2^2$).
- 3 Alignment Check? The first network's 3rd octet is **4**. Is 4 divisible by the number of networks (4)? Yes ($4 \bmod 4 == 0$).

- **The Aggregation Math:**

- We are combining $4 = 2^2$ networks. We move the prefix **left by 2 bits**.
- Old Prefix: /24. New Prefix: $24 - 2 =$ **/22**.
- **Final Supernet Route:** 200.1.4.0 /22.

GATE Example: A Failing Supernet

- **Given Networks:**

- 200.1.5.0 /24
- 200.1.6.0 /24
- 200.1.7.0 /24
- 200.1.8.0 /24

- **Applying the Rules:**

- ① Contiguous? Yes (5, 6, 7, 8).
- ② Power of 2? Yes ($4 = 2^2$).
- ③ Alignment Check? The first network's 3rd octet is **5**. Is 5 divisible by 4? **No**.

- **Conclusion:** These 4 networks **cannot** be combined into a single /22 supernet.
- *Why?* Because mathematically, the bitwise boundary of a /22 block starting at .5 would actually span IPs outside this range, bringing in foreign networks and causing routing loops!